

Valuation Project 3: Monte Carlo Methods

Choice #5 Lookback option with a simple payoff

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1. Executive Summary

Using a Monte Carlo simulation with $100 \times 1,000,000$ paths under the risk-neutral measure, we estimate the fair value of the security at **\$979.62** per note (SE = 0.016), deviating +0.15% from the issuer's market estimated value of \$978.20.

2. Parameter Selection

N (Simulation paths)	1,000,000	K (Strike level)	100%, 80%
T (Time)	790 days (1/9/26 ~ 3/14/28)	r (Risk-free rate)	3.2625% 3.3456%
M (Time steps)	24	σ (Implied volatility)	19.146%
S_0 (Initial level)	\$6,966.28 (Subject to change)	q (Dividend yield)	1.186%

• N (Simulation paths) - 1,000,000

We chose $N = 1,000,000$ for our simulation to ensure that the Standard Error is minimized to a negligible level. Given the path-dependent nature of the lookback option, a large sample size is required to accurately capture the distribution of asset price minimums. The resulting tight range of our 100 trial results with a standard error of 0.01632 proves that 1,000,000 simulations provide a highly stable and reliable valuation.

• M (Time steps) - 24

In our Monte Carlo simulation, $M = 24$ represents the total number of time steps used to discretize the lifespan of the product.

A key mathematical advantage of the Monte Carlo method is its flexibility regarding the time grid. The time steps do not need to be uniform in length. Instead of simulating every single day over the entire life of the option, which would be computationally expensive. We vary the step size based on the observation frequency required by the product's terms.

We break down the 24 time steps as follows:

- Time Steps 1 to 22 (Initial Observation Period): Each of the first 22 steps has a uniform length of exactly 1 day ($dt = 1/365$). This matches the Initial Observation Period from 1/9/26 to 1/30/26. Daily simulation during this specific window is required because the initial strike or reference level depends on the lowest closing price observed across these 22 days.

- Time Step 23 (First Determination Period): The 23rd time step transitions to a much larger interval, spanning 403 days from 1/30/26 to 3/9/27 ($dt = 403/365$). Because the product only checks for early redemption on 3/9/27, simulating the daily asset paths between these two dates is unnecessary. A single, long time step efficiently evolves the asset price to the first determination date.
- Time Step 24 (Final Determination Period): The final time step spans 366 days from 3/9/27, to 3/9/28 ($dt = 366/365$). If the autocall condition was not triggered at step 23, the model advances the price path to the final determination date to evaluate the payoff at maturity.

- **T (Time) - 790 days (1/9/26 ~ 3/14/28)**

To ensure rigorous valuation, we explicitly defined seven key dates based on the product prospectus.

Following a discussion with the professor, we explicitly set the model's origin T_0 to 1/9/26 (the inception of the lookback period) to directly capture path-dependent realizations without introducing any structural distortion.

Exact Minimum Capture: Because the initial level is defined as the lowest closing level observed between 1/9/2026 and 1/30/2026, initiating the simulation paths at T_0 ensures that every simulated daily closing price during the lookback period is precisely evaluated. We can calculate the exact minimum for each of the simulated paths without requiring any interpolation or numerical smoothing.

Unified Timeline Framework: Aligning T_0 with the contract's start date keeps both the daily lookback period and the long-term determination periods within a single, unified simulation framework. This eliminates any temporal mismatch and maximizes the accuracy of the path-dependent payoff calculations.

- T_0 : 1/9/26 Pricing date (The market data is observed and the contract terms are struck.)
- T_1 : 1/22/26 Issue date (The securities are issued and initial funds are exchanged.)
- T_2 : 1/30/26 End of Initial observation period (The initial level will be defined as the lowest closing level from 1/9/26 to 1/30/26.)
- T_3 : 3/9/27 First determination date (The securities will be automatically redeemed if the closing level of the underlier is greater than or equal to the call threshold level)
- T_4 : 3/12/27 Early redemption date (The early redemption payment will be made if the securities have not been automatically redeemed prior to maturity.)
- T_5 : 3/9/28 Final determination date (The underlying asset's final closing price ($S_\square$) is recorded to determine the payoff.)
- T_6 : 3/14/28 Maturity date (The final cash flows are actually paid to investors if the securities have not been automatically redeemed prior to maturity.)

Our Monte Carlo simulation divides the lifespan of the security into three distinct intervals to maximize computational efficiency while fully capturing all contract features.

1. Initial Observation Period $T_2 - T_0$ (22 days, or 0.0603 year)

- Mechanism: The model simulates the underlying price path day-by-day for the first 22 days.
- Payoff Impact: The lowest closing level observed during this specific 22-day period is recorded as the initial level and strike price for that individual path. To comply with the prospectus, this initial level is capped at the closing price observed on 1/9/26 (\$6,966.28).

2. First Determination Period $T_3 - T_0$ (424 days, or 1.1616 years)

- Mechanism: Using a single time step from T_2 to T_3 (403 days), the model projects the asset's price directly to the first determination date.
- Payoff Impact: If the simulated price at T_3 meets or exceeds the autocall threshold level, the security triggers early redemption. The payoff is calculated, discounted back to T_0 , and the simulation path terminates.

3. Final Observation Period $T_5 - T_0$ (790 days, or 2.1644 years)

- Mechanism: For paths that did not trigger early redemption at T_3 , the model uses a final step of 366 days to project the asset price from T_3 to the final determination date (T_5).
- Payoff Impact: This final price determines whether the investor receives the original principal, a positive performance return, or faces downside risk based on the lookback strike generated during the first period.

● **S_0 (Initial level) - 6,966.28 (Subject to change)**

According to the term sheet, the initial level will be defined as the lowest closing level during the initial observation period (1/9/26 ~ 1/30/26). The closing level on 1/9/26 is 6,966.28.

● **K (Strike level) - 100%, 80%**

For the automatic early redemption, the strike level is 100% of the initial level. The securities will be automatically redeemed if the closing level of the underlier is greater than or equal to the call threshold level on the first determination date for the early redemption payment.

If the securities have not been redeemed, the payment at maturity will be made. If the final level is greater than 100% of the initial level, investors will receive the stated principal (\$1000) amount plus the upside payment.

If the final level is equal to or less than the initial level but is greater than or equal to the downside threshold level (80% of the initial level), investors will receive only the stated principal amount at maturity.

If, however, the final level is less than the downside threshold level, investors will lose 1% for every 1% decline in the level of the underlier over the term of the securities.

● **r (Risk-free rate) - 3.2625%, 3.3456%**

We obtained the risk-free rate and discount factor data from Bloomberg and interpolated the term structure to match the maturity of the security. We searched <USD OIS> on Bloomberg and set the date to 1/7/26 which was two days before the pricing date (1/9/26) due to swap settlement time.

Using OIS for the risk-free rate is correct because it reflects the overnight indexed swap rate, which serves as the modern market standard for a "risk-free" benchmark by minimizing credit and liquidity risk premiums. Since these structured products are often collateralized under Credit Support Annex (CSA) agreements, OIS accurately represents the actual funding and discounting costs used by financial institutions. Incorporating a precise OIS-based discount factor ensures the valuation model remains consistent with non-arbitrage pricing principles and market hedging practices.

We performed linear interpolation on Bloomberg Discount Factors (DF) to match our model's key dates and then converted these into continuously compounded risk-free rates (r) using the formula: $r = -\ln(\text{DF}) / T$.

1. Maturity Discount Rate ($r = 3.2625\%$):

- This rate is used to discount the final payoff at the end of the 790-day observation period (T_5).
- Using Bloomberg DF quotes for 11/2/28 (0.936750) and 11/2/29 (0.906261), we interpolated to find the DF for the target maturity date, resulting in a 3.2625% zero rate.

2. Early Redemption Discount Rate ($r = 3.3456\%$):

- This rate is applied if the autocall condition is met on the first determination date (T_3).
- Using Bloomberg DF quotes for 1/11/27 (0.966509) and 7/9/27 (0.951913), we interpolated to find the DF for the early redemption date (T_4), resulting in a 3.3456% zero rate.

This dual-rate approach allows our model to maintain a locally flat term structure for each potential cash flow event, isolating the S&P 500's price dynamics while ensuring the valuation remains consistent with the market's yield curve and non-arbitrage pricing principles.

● **σ (Implied volatility) - 19.146%**

We obtained the implied volatility data for the S&P 500 Index from Bloomberg <OVM>. To accurately reflect market expectations over the lifespan of the security, we selected the 790-day at-the-money (100% moneyness) implied volatility, which stands at 19.146%.

While implied volatility varies across different tenors and strike prices in the market (the volatility surface), we intentionally select $\sigma = 19.146\%$ as the constant volatility parameter for our Monte Carlo simulation for the following reasons:

- Alignment with the Black-Scholes GBM Framework: The standard Monte Carlo valuation model assumes that the underlying asset price follows a Geometric Brownian Motion (GBM) with constant volatility. To maintain consistency within this non-arbitrage framework, a single volatility input is required.

- **Maturity Matching Principle:** The security has a maximum lifespan of 790 days ($T = 2.1644$ years). Since the terminal payoff and long-term asset dynamics are the primary drivers of the instrument's overall value, selecting the volatility that matches the 790-day horizon ensures that the long-term diffusion of the simulated asset paths is correctly calibrated to current market prices.
- **Simplification of the Volatility Term Structure:** Although the security includes an initial 22-day lookback period and a 424-day early redemption observation period, incorporating a time-varying or local volatility model would introduce excessive computational complexity and calibration risk. By utilizing the 790-day at-the-money implied volatility, we effectively capture the average expected variance over the entire life of the option, striking an optimal balance between valuation accuracy and computational efficiency.

- **q (Dividend yield) - 1.186%**

The dividend yield (1.186%) of the underlying S&P 500 index was obtained from Bloomberg <OVME>. Since the underlying asset is a broad market index, applying a continuous dividend rate is more appropriate and computationally efficient for the Monte Carlo method than modeling discrete dividend payments.

3. Algorithm Description

3.1 Model Setup

Under the risk-neutral measure $\mathbb{Q}$, the underlying SPX index follows a Geometric Brownian Motion. For each simulated path n , the price evolves over a discrete time grid $\{T_0, T_1, \dots, T_{24}\}$ according to:

$$S^n_{T_i} = S^n_{T_{i-1}} \cdot \exp[(r - q - \frac{1}{2}\sigma^2) \cdot \Delta t_i + \sigma \cdot \sqrt{\Delta t_i} \cdot \varphi^n_i] \text{ where } \Delta t_i = T_i - T_{i-1}$$

$\varphi^n_i \sim N(0, 1)$ i.i.d. across paths and time steps

$n = 1, 2, \dots, N$ (path index)

This recursion is the exact closed-form solution of GBM

3.2 Time Grid

The time nodes are anchored at $t = 0$ (January 9, 2026, $S_0 = 6,966.28$) and aligned to the contractual observation dates:

$T_0 = 0$ (pricing date)

$T_1 = 1 / 365$ (day 1)

$T_2 = 2 / 365$ (day 2)

⋮

$T_{22} = 21 / 365$ (Initial Observation ends, January 30, 2026)

$$T_{23} = 424 / 365 \quad (\text{First Determination, March 9, 2027})$$

$$T_{24} = 790 / 365 \quad (\text{Final Determination, March 9, 2028})$$

The first 22 intervals ($T_0 \rightarrow T_{22}$) use daily steps ($\Delta t_i = 1/365$) to capture the running minimum. The intervals $[T_{22}, T_{23}]$ and $[T_{23}, T_{24}]$ are simulated as single large time steps (403 and 366 days respectively), exploiting the independent-increments property of Brownian motion. This reduces the per-path step count from 790 to 23 with no approximation error, since the GBM solution is exact for any time interval.

3.3 Stage 1 — Initial Level (Lookback Strike)

For each path n , the initial level is determined as the running minimum of the underlier over the initial observation period:

$$MIN^n = \min (S^n_{\{T_1\}}, S^n_{\{T_2\}}, \dots, S^n_{\{T_{22}\}})$$

This MIN^n serves as the strike level K^n for both the autocall barrier and the terminal payoff conditions on path n . Per contract, the initial level is also capped at $S_0 = 6,966.28$; this cap is enforced implicitly since $S^n_{\{T_0\}} = S_0$ anchors the path.

3.4 Stage 2 — Autocall Observation at T_{23}

At T_{23} (March 9, 2027), the path bifurcates into two scenarios:

Scenario A — Autocall triggered ($S^n_{\{T_{23}\}} \geq MIN^n$):

The path receives the early redemption payment, settled three calendar days later:

$$V^n_{\{T_{23}\}} = \$1,098.50$$

$$PV^n = \$1,098.50 \cdot \exp(-r \cdot (T_{23} + 3/365))$$

The contract terminates here; no further cash flows occur on path n .

Scenario B — Autocall not triggered ($S^n_{\{T_{23}\}} < MIN^n$):

The path continues to maturity at T_{24} . The cash flow is determined in Stage 3.

3.5 Stage 3 — Maturity Payoff at T_{24} (Conditional on Scenario B)

At T_{24} (March 9, 2028), the terminal payoff is a piecewise function of the final level $S^n_{\{T_{24}\}}$ relative to MIN^n , decomposed into three sub-scenarios:

Scenario B1 — Upside participation ($S^n_{\{T_{24}\}} > MIN^n$):

$$V^n_{\{T_{24}\}} = \$1,000 + 1.5 \cdot \$1,000 \cdot (S^n_{\{T_{24}\}} / MIN^n - 1)$$

The investor receives the full principal plus 150% leveraged participation in the underlier's appreciation.

Scenario B2 — Capital protection ($0.8 \cdot MIN^n \leq S^n_{\{T_{24}\}} \leq MIN^n$):

$$V^n_{T_{24}} = \$1,000$$

The investor receives the full principal; downside up to 20% is fully absorbed.

Scenario B3 — Knock-in loss ($S^n_{T_{24}} < 0.8 \cdot MIN^n$):

$$V^n_{T_{24}} = \$1,000 \cdot (S^n_{T_{24}} / MIN^n)$$

The investor incurs a 1-for-1 loss linked to the underlier's decline relative to MIN^n . The payment can be as low as zero.

For Scenarios B1, B2, and B3, the maturity cash flow is paid five calendar days after T_{24} :

$$PV^n = V^n_{T_{24}} \cdot \exp(-r \cdot (T_{24} + 5/365))$$

3.6 Monte Carlo Estimator

The fair value estimate is the sample mean of path-wise present values:

$$\hat{V}_0 = (1/N) \cdot \sum_{n=1}^N PV^n$$

with standard error

$$SE = s_{PV} / \sqrt{N} \quad \text{where} \quad s_{PV}^2 = (1/(N-1)) \cdot \sum_{n=1}^N (PV^n - \hat{V}_0)^2$$

The 95% confidence interval is $\hat{V}_0 \pm 1.96 \cdot SE$.

4. Model Robustness Test and Sensitivity Analysis

4.1 Robustness Test of N

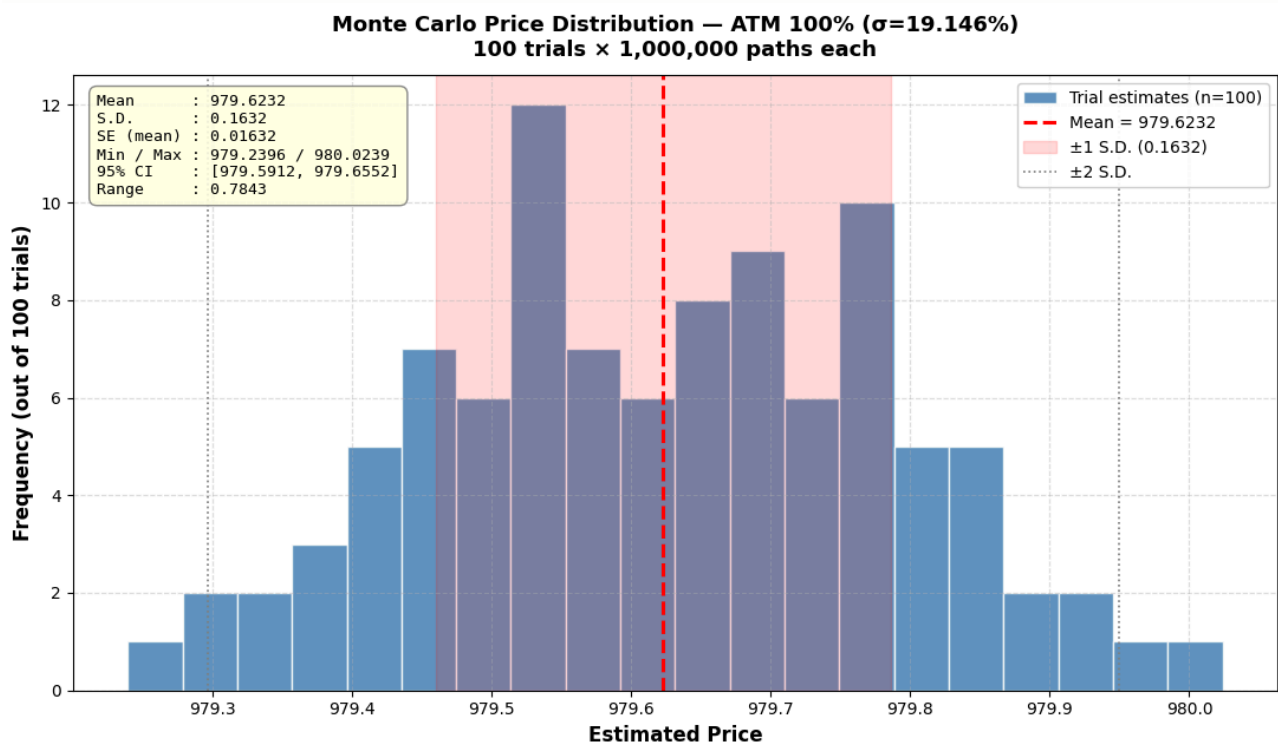


Figure 1. Distribution of fair value estimates across 100 independent Monte Carlo trials, each comprising 1,000,000 paths. The dashed red line marks the sample mean (\$979.62); the shaded region indicates ± 1 standard deviation (\$979.46 ~ \$979.78).

The trial-to-trial standard deviation of 0.16 yields a final standard error of 0.016 — a relative precision of 0.0017% of price. The near-normal distribution empirically validates the Central Limit Theorem.

4.2 Robustness of Possible Error

- Statistical Sampling Error: Scales as $1/\sqrt{N}$. At $N = 10^8$, reduced to 0.016, well below typical bid-ask spreads for institutional structured notes.
- Flat Term Structure within Each Horizon: Our dual-rate discounting (3.2625% / 3.3456%) handles the term structure between the two termination dates but assumes flatness within each horizon.

4.3 Sensitivity Analysis of Volatility

While our base case uses the 790-day ATM (100% moneyness) implied volatility ($\sigma = 19.146\%$) to match the contract's full maturity horizon, the implied volatility surface contains four points relevant to this product. Table X compares the MC fair value under each:

IV Scenario	σ	MC Fair Value	Deviation from \$978.20
ATM 100% / 790-day (Base Case)	19.146%	\$979.62	+0.15%
Downside 80% / 790-day	23.944%	\$960.30	-1.83%
ATM / 424-day (autocall horizon)	17.770%	\$985.30	+0.73%
ATM / 21-day (lookback horizon)	11.307%	\$1,011.38	+3.39%

Each volatility input probes a different product feature: the 80% IV reflects pricing in the knock-in barrier region; the 424-day IV reflects autocall probability; the 21-day IV reflects the lookback observation period.

The ATM 790-day IV yields by far the smallest deviation from the issuer's estimated value (+0.15%), strongly suggesting the issuer's internal model is anchored to a maturity-matched ATM volatility. The full spread across the four IV inputs spans \$51 per note (~5.1% of price), which dominates every other source of error in this valuation: the lookback 21-day IV (\$1,011) inflates the price because low volatility prevents the strike K from being depressed, while the downside 80% IV (\$960) deflates it by amplifying knock-in losses. This empirical evidence demonstrates that the

volatility surface choice is the single largest source of model risk for this product — far exceeding the Monte Carlo statistical error (0.016) or the discretization bias relative to CRR (~\$1).

4.4 Comparison with Binomial Tree Model

We previously implemented a Binomial Tree model which is more complicated. While that model required significant structural engineering to align nodes with path-dependent features, we have transitioned to the Monte Carlo method for this specific valuation to better handle the continuous nature of the lookback feature.

The following table summarizes the performance and valuation results for the financial instrument (Actual Market Price: \$978.20):

Feature	Binomial Tree Model	Monte Carlo Simulation
Estimated Price	\$980.68	\$979.62
Execution Time	2.036 seconds	108.80 seconds
Error Type	Non-linearity Error	Random Sampling Error
Pricing Error	~0.25%	~0.15%
Reduces with N	No (Bias drifts)	Yes ($SE \propto 1/\sqrt{N}$)

While the Binomial Tree offers superior speed for simpler derivatives, the Monte Carlo method provides the necessary flexibility and robustness for the path-dependent features of our lookback option. The Monte Carlo method is significantly easier to implement for complex payoffs, as it avoids the intricate grid-alignment and node-tracking logic required by a tree. Despite the longer execution time, the Monte Carlo model's ability to handle non-uniform time steps and exact daily observations makes it the more reliable framework for valuing sophisticated structural products.

5. Final Recommendation

Our final estimated value is \$979.62 per note by the Monte Carlo valuation.

Our analysis confirms that the Monte Carlo method with N=1,000,000 simulations provides a precise and robust framework for this path-dependent security. We recommend utilizing the ATM 100% Moneyness (19.146%) implied volatility as the primary input, as it minimizes pricing error relative to the market benchmark of \$978.20 and exhibits stability when capturing the lookback feature and autocall probabilities. By incorporating a dual-rate discounting framework (3.2625% and 3.3456%), the model remains consistent with non-arbitrage pricing principles and reflects the actual funding costs associated with the security's different potential termination horizons.